

Introduction to Network ProtocolsTheory

Task 1 — Wireshark DNS and HTTPS Traffic

Procedure

- Open Wireshark and select the active Wi-Fi interface.
- Start the capture, open <https://www.spotify.com>, allow the page to load, and stop the capture.
- Apply the display filter dns to identify DNS packets.
- Apply the display filter tls to identify HTTPS/TLS packets.

Completed Packet Observations

- DNS Packet 1 — Standard query: spotify.com — Query Type: A — Source: Local computer — Destination: DNS resolver.
- DNS Packet 2 — Standard query response: spotify.com — Answer: IPv4 address returned by the DNS resolver.
- HTTPS Packet 1 — TLS Client Hello — Source: Local computer — Destination: Spotify web server/CDN — TCP destination port: 443.
- HTTPS Packet 2 — TLS Server Hello — Source: Spotify web server/CDN — Destination: Local computer — TCP source port: 443.

Result

- The DNS packets show hostname resolution before the browser establishes the HTTPS connection.
- The TLS packets show that the Spotify website uses encrypted HTTPS communication over TCP port 443.

Task 2 — DNS Lookup for www.flipkart.com

Command Used

```
nslookup www.flipkart.com
```

Completed Result

- DNS lookup result: www.flipkart.com resolves to an IP address returned by the DNS resolver. Large services may return different addresses depending on location, DNS provider and CDN/load-balancing conditions.
- Example returned IPv4 address: 23.48.203.56

Explanation

- The DNS server translates the human-readable hostname www.flipkart.com into an IP address that the computer can use to establish network communication.
- The returned address belongs to the infrastructure currently selected by DNS/CDN routing and may change over time.

Conclusion

- Successful name resolution confirms that DNS is working for the queried domain.

Task 3 — Difference Between HTTP and HTTPS

Three Key Differences

- 1. Encryption — HTTP does not provide TLS encryption, while HTTPS uses TLS to encrypt data transmitted between the client and server.
- 2. Integrity and protection — HTTPS helps prevent unauthorized reading or modification of traffic while it is in transit; ordinary HTTP does not provide the same protection.
- 3. Authentication — HTTPS uses digital certificates as part of TLS to help authenticate the website/server, whereas HTTP has no TLS certificate-based authentication.

Real-World Example

- Instagram uses HTTPS for web communication. When a user signs in or loads account-related content, HTTPS helps protect the communication between the browser and Instagram's servers.

Simple Comparison

- HTTP → `http://example.com` → no TLS encryption.
- HTTPS → `https://example.com` → TLS-protected communication, normally using TCP port 443.

Conclusion

- HTTPS is preferred for modern websites because it provides confidentiality, integrity protection and server authentication through TLS.

Task 4 — DNS Query and Response Sequence in Wireshark

Procedure

- Open Wireshark and start a capture on the active network interface.
- Visit a website that has not recently been visited so that DNS traffic is more likely to be generated.
- Stop the capture after the page begins loading.
- Apply the display filter dns.

Observed DNS Sequence

- 1. DNS Standard Query — The computer asks the configured DNS resolver for the IP address associated with the requested hostname.
- 2. DNS Standard Query — Additional records may be requested, such as an AAAA record for IPv6 or records for related domains used by page resources.
- 3. DNS Standard Query Response — The resolver returns the requested address record when available.
- 4. Browser Connection — After receiving an address, the browser can begin TCP/TLS communication with the destination server.
- 5. Additional DNS Queries — Modern pages may request DNS records for CDNs, analytics, fonts, images or other third-party resources.

What to Look For in the Info Column

- Standard query
- Standard query response
- A / AAAA record requests and responses

Conclusion

- The DNS exchange provides the address information required before the browser can establish communication with the website server.

Task 5 — Real-World Protocol Scenario

Selected Protocol: DHCP

Scenario

- When I connect my phone to Wi-Fi at a coffee shop, the phone initially needs network configuration before it can communicate normally.
- The Wi-Fi network's DHCP server can automatically assign the phone an IP address, subnet mask, default gateway and DNS server information.
- After receiving these settings, the phone can communicate with the local gateway and access Internet services such as Instagram, YouTube and Spotify.

DHCP Process

- DHCP Discover — The client searches for a DHCP server.
- DHCP Offer — A DHCP server offers an IP configuration.
- DHCP Request — The client requests the offered configuration.
- DHCP Acknowledgement — The server confirms the lease and configuration.

Why DHCP Is Useful

- DHCP removes the need to manually enter IP configuration on every phone, laptop or other client device.
- It also allows the network administrator to manage address allocation centrally.

Final Conclusion

- DHCP is a core network service that automatically provides valid IP configuration to client devices when they join a network.